

Renewable Energy Orchestrator — Next Steps

Oct 3, 2026 · @RIBHU Dubey

Covers what's left before the hackathon submission: the Learning Guide research, remaining MVP verification, known code gaps, and the final submission checklist.

Domain Grounding (Learning Guide)

The team's Learning Guide lays out a 2-3 hour path to ground the Evaluator's rules in real grid-operator principles, not guesses:

- ☐ Step 1 (2-3 hrs): research merit order dispatch, battery arbitrage, curtailment causes, grid frequency/reliability basics
- ☐ Step 2: refine the 6 hard-fail rules against what you learn — especially Rule 3 (reliability) and Rule 5 (the daily-price-high threshold is currently a crude placeholder)
- ☐ Step 3: paste the refined checklist + objective logic into a fresh LLM conversation and ask it to sanity-check as a grid-ops expert
- ☐ Step 4: revisit the per-objective "optimal" definitions once Step 1 is done (the objective structure itself is already built)
- ☐ Step 5 (optional, 15 min): show 3-4 real decisions from the running system to anyone with adjacent energy/trading experience

Verify the MVP End-to-End

- ☒ ~~Scenario generation confirmed across difficulties, including multi_failure_cascade~~
- ☒ ~~Orchestrator Agent returns a valid decision via Gemini, with reasoning that correctly applies merit order, battery logic, curtailment as last resort, and reliability~~
- ☒ ~~Evaluator correctly flags a reliability violation on a real decision (caught the orchestrator under counting industrial demand)~~
- ☐ Batch Run panel not yet tested with multiple scenarios per difficulty
- ☐ "Strip objective" fallback behavior not yet confirmed in a batch run (only confirmed in single-scenario testing)

Known Gaps Before Submission

Gap	Where	Priority
<code>rule_5</code> daily-high threshold is a placeholder, not a real daily price curve	<code>backend/app/data/rules.py</code>	Medium
<code>max_profit</code> objective check always passes — no real check implemented	<code>backend/app/agents/evaluator.py</code>	Medium
No time-series simulation — scenarios are single independent ticks, needed to claim F3	Scenario Agent	High
No multimodal input — structured JSON only; decide whether to pursue D3 before investing here	Scenario Agent	Low
Orchestrator could still schedule an action against an offline battery — not cross-checked	Evaluator	Medium
No persistence — every run is stateless/in-memory	Backend	Low

Fixed since the MVP was first built: Gemini free-tier `503` "model overloaded" errors on `gemini-3.8-flash` — the Orchestrator now retries automatically, and the default model was switched to `gemini-3.5-flash-lite`.

Before Final Submission

Per the hackathon’s own requirements, the submission needs two components together showing what the solution does, how it works, and how the claimed solution features and depth are demonstrated:

- ☐ A working demo covering every area claimed — script the 3-beat narrative: bulk auto run (pass-rate at scale) → live manual edge case → stripped-objective fallback
- ☐ Detailed structural architecture — the Architecture & Features document covers this; update its 9-blocker table once the actual F/D coverage is final
- ☐ Self-declared 9-blocker position matches the demo evidence exactly — the hackathon penalizes over-estimation and under-estimation equally